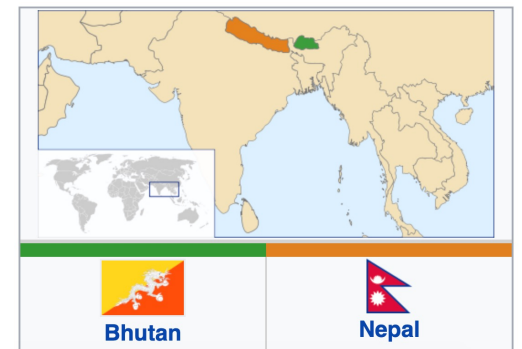


20: Maximum Likelihood Estimation

Jerry Cain
May 15th, 2024

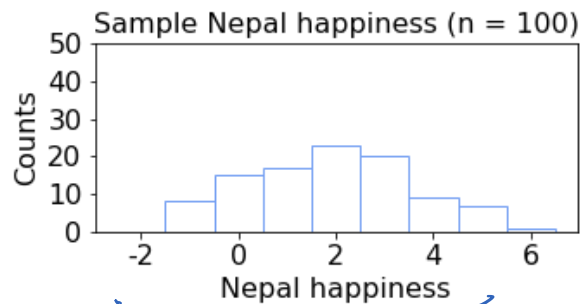
[Lecture Discussion on Ed](#)

Bootstrap: p-value



Null hypothesis test

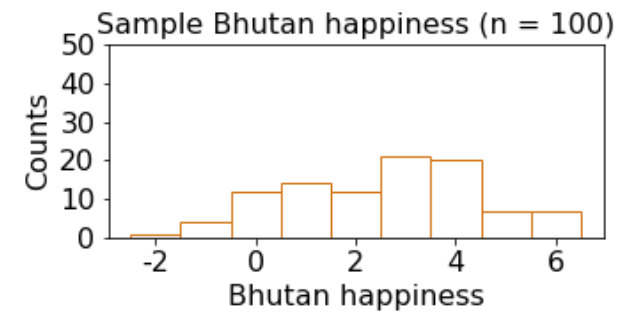
Nepal Happiness
4.45
2.45
6.37
2.07
...
1.63



note happiness scaling is different

$$\bar{X}_1 = 3.1$$

Bhutan Happiness
0.91
0.34
1.91
1.61
...
1.08



$$\bar{X}_2 = 2.4$$

Claim: The difference in mean happiness between Nepal and Bhutan is 0.7 happiness points, and **this is statistically significant.**

Null hypothesis test

def **null hypothesis** – It's possible the two distributions are essentially the same, and your claim looks significant but really is not.

def **p-value** – What is the probability that the observed difference occurs under the null hypothesis? *restated: is there a chance that you observed a significant difference by chance, but it's far from reproducible*

Example:

- Flip some coin 100 times.
- Flip some coin 150 times.
- Compute fraction of heads in both groups.
- There is a possibility we'll see the observed difference in these fractions even if we used the same coin

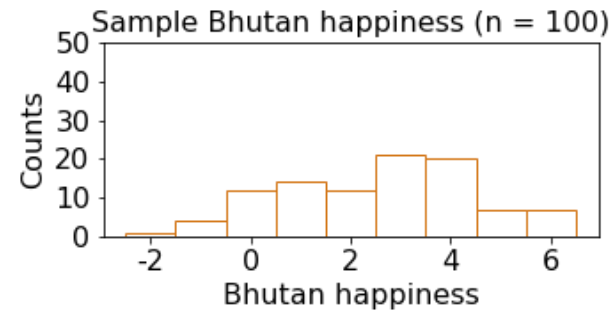
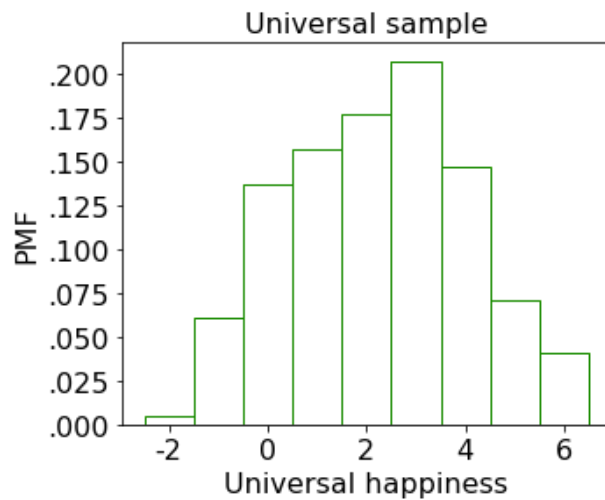
} **Null hypothesis** assumes we use the same coin

} **p-value**

A **significant** p-value (< 0.05) means we reject the null hypothesis.

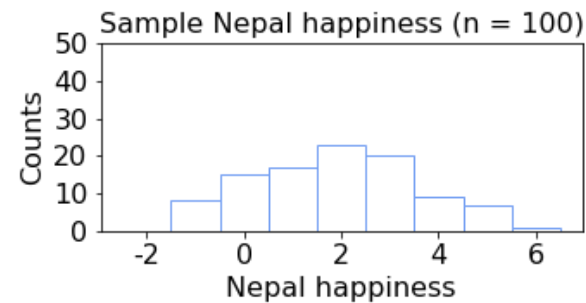
Universal sample

(this is what the null hypothesis assumes)



$$\bar{X}_1 = 3.1$$

how often do we see this level of difference from the combined population



$$\bar{X}_2 = 2.4$$

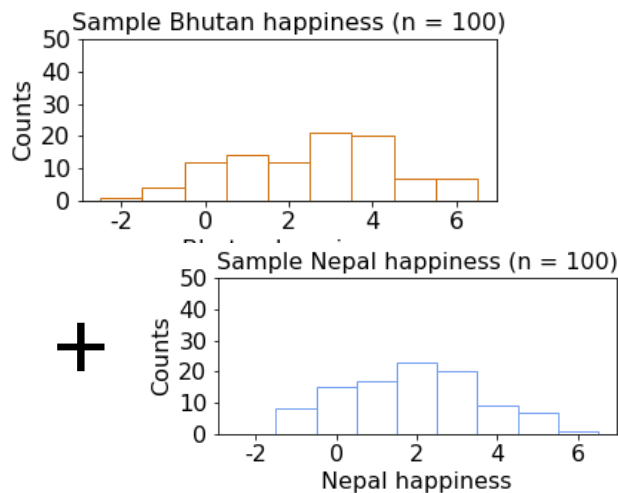
Want **p-value**: probability $|\bar{X}_1 - \bar{X}_2| \geq |3.1 - 2.4|$ happens under null hypothesis

Bootstrap for p-values

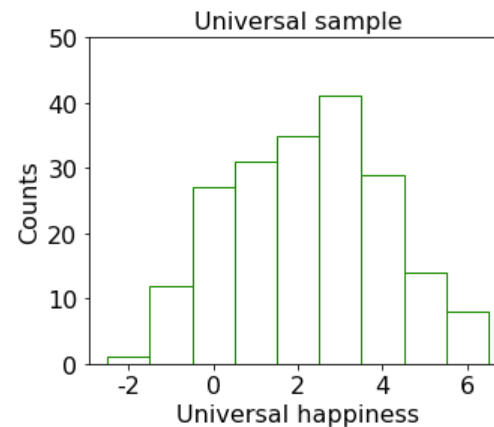
if two statistics were only very different by chance, then combining the two populations should create a population that's essentially the same as the two originals.

1. Create a **universal sample** using your two samples

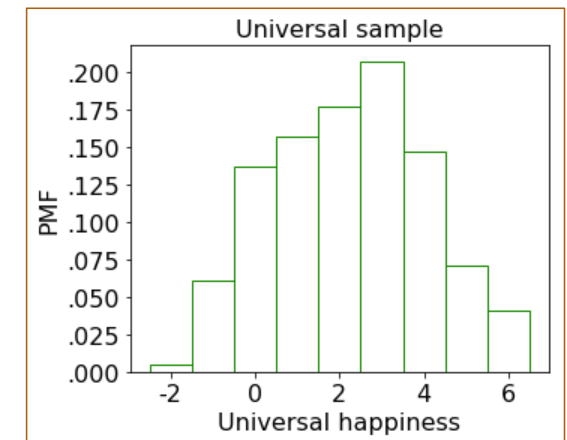
i.e., recreate the null hypothesis



=



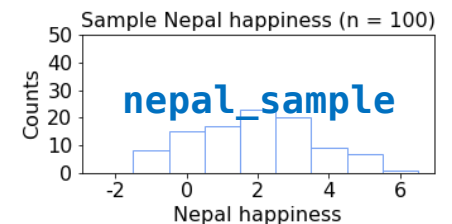
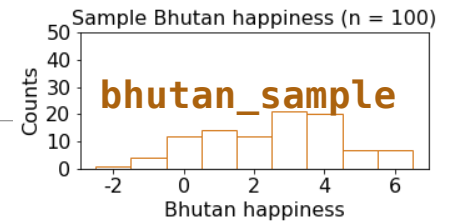
≈



Bootstrap for p-values

1. Create a **universal sample** using your two samples
2. Repeat **10,000** times:
 - a. Resample **both samples**
 - b. Recalculate the **mean difference** between the resamples
3. **p-value** =
$$\frac{\# (\text{mean diffs} \geq \text{observed diff})}{n = 10000}$$

this number doesn't have to be 10000, but it needs to be large.



Probability
that observed
difference arose
by chance

Bootstrap for p-values

```
def pvalue_boot(bhutan_sample, nepal_sample):  
    N = size of the bhutan_sample  
    M = size of the nepal_sample  
    observed_diff = |mean of bhutan_sample – mean of nepal_sample|  
  
    uni_sample = combine bhutan_sample and nepal_sample  
    count = 0  
  
    repeat 10,000 times:  
        bhutan_resample = draw N resamples from the uni_sample  
        nepal_resample = draw M resamples from the uni_sample  
        muBhutan = sample mean of the bhutan_resample  
        muNepal = sample mean of the nepal_resample  
        diff = |muNepal – muBhutan|  
        if diff >= observed_diff:  
            count += 1  
  
pValue = count / 10,000
```

Bootstrap for p-values

1. Create a universal sample using your two samples

```
def pvalue_boot(bhutan_sample, nepal_sample):  
    N = size of the bhutan_sample  
    M = size of the nepal_sample  
    observed_diff = |mean of bhutan_sample – mean of nepal_sample|  
  
    uni_sample = combine bhutan_sample and nepal_sample  
    count = 0
```

repeat 10,000 times:

```
    bhutan_resample = draw N resamples from the uni_sample  
    nepal_resample = draw M resamples from the uni_sample  
    muBhutan = sample mean of the bhutan_resample  
    muNepal = sample mean of the nepal_resample  
    diff = |muNepal – muBhutan|  
    if diff >= observed_diff:  
        count += 1
```

pValue = count / 10,000

Bootstrap for p-values

2. a. Resample both samples

```
def pvalue_boot(bhutan_sample, nepal_sample):  
    N = size of the bhutan_sample  
    M = size of the nepal_sample  
    observed_diff = |mean of bhutan_sample - mean of nepal_sample|  
  
    uni_sample = combine bhutan_sample and nepal_sample  
    count = 0  
  
    repeat 10,000 times:  
        bhutan_resample = draw N resamples from the uni_sample  
        nepal_resample = draw M resamples from the uni_sample  
        muBhutan = sample mean of the bhutan_resample  
        muNepal = sample mean of the nepal_resample  
        diff = |muNepal - muBhutan|  
        if diff >= observed_diff:  
            count += 1  
  
    pValue = count / 10,000
```

it's important that you go with precisely the same sizes, else you're using different random variables.

Bootstrap for p-values

2. b. Recalculate the **mean difference** b/t resamples

```
def pvalue_boot(bhutan_sample, nepal_sample):  
    N = size of the bhutan_sample  
    M = size of the nepal_sample  
    observed_diff = |mean of bhutan_sample – mean of nepal_sample|  
  
    uni_sample = combine bhutan_sample and nepal_sample  
    count = 0  
  
    repeat 10,000 times:  
        bhutan_resample = draw N resamples from the uni_sample  
        nepal_resample = draw M resamples from the uni_sample  
        muBhutan = sample mean of the bhutan_resample  
        muNepal = sample mean of the nepal_resample  
        diff = |muNepal – muBhutan|  
        if diff >= observed_diff:  
            count += 1
```

pValue = count / 10,000

Bootstrap for p-values

$$3. \text{ p-value} = \frac{\# (\text{mean diffs} > \text{observed diff})}{n}$$

```
def pvalue_boot(bhutan_sample, nepal_sample):  
    N = size of the bhutan_sample  
    M = size of the nepal_sample  
    observed_diff = |mean of bhutan_sample – mean of nepal_sample|  
  
    uni_sample = combine bhutan_sample and nepal_sample  
    count = 0  
  
    repeat 10,000 times:  
        bhutan_resample = draw N resamples from the uni_sample  
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        muBhutan = sample mean of the bhutan_resample  
        muNepal = sample mean of the nepal_resample  
        diff = |muNepal – muBhutan|  
        if diff >= observed_diff:  
            count += 1
```

pValue = count / 10,000

Lisa Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Spring 2024

Bootstrap for p-values

```
def pvalue_boot(bhutan_sample, nepal_sample):  
    N = size of the bhutan_sample  
    M = size of the nepal_sample  
    observed_diff = |mean of bhutan_sample – mean of nepal_sample|
```

```
    uni_sample = combine bhutan_sample and nepal_sample  
    count = 0
```

```
    repeat 10,000 times:
```

with replacement!

```
        bhutan_resample = draw N resamples from the uni_sample  
        nepal_resample = draw M resamples from the uni_sample  
        muBhutan = sample mean of the bhutan_resample  
        muNepal = sample mean of the nepal_resample  
        diff = |muNepal – muBhutan|  
        if diff >= observed_diff:  
            count += 1
```

```
pValue = count / 10,000
```

Bootstrap



Let's go try it!

Google colab notebook [link](#)

Null hypothesis test

Nepal
Happiness

4.45

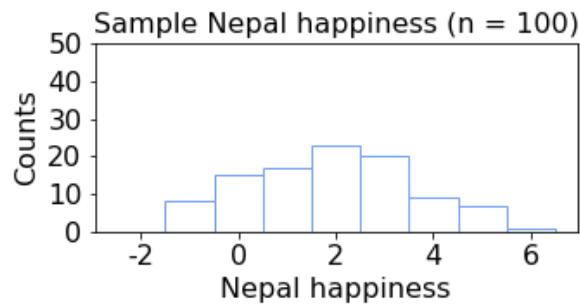
2.45

6.37

2.07

...

1.63



$$\bar{X}_1 = 3.1$$

Bhutan
Happiness

0.91

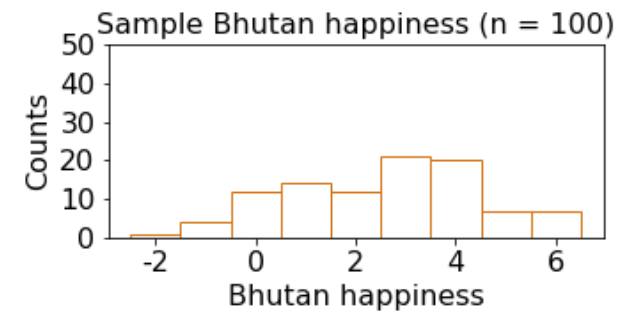
0.34

1.91

1.61

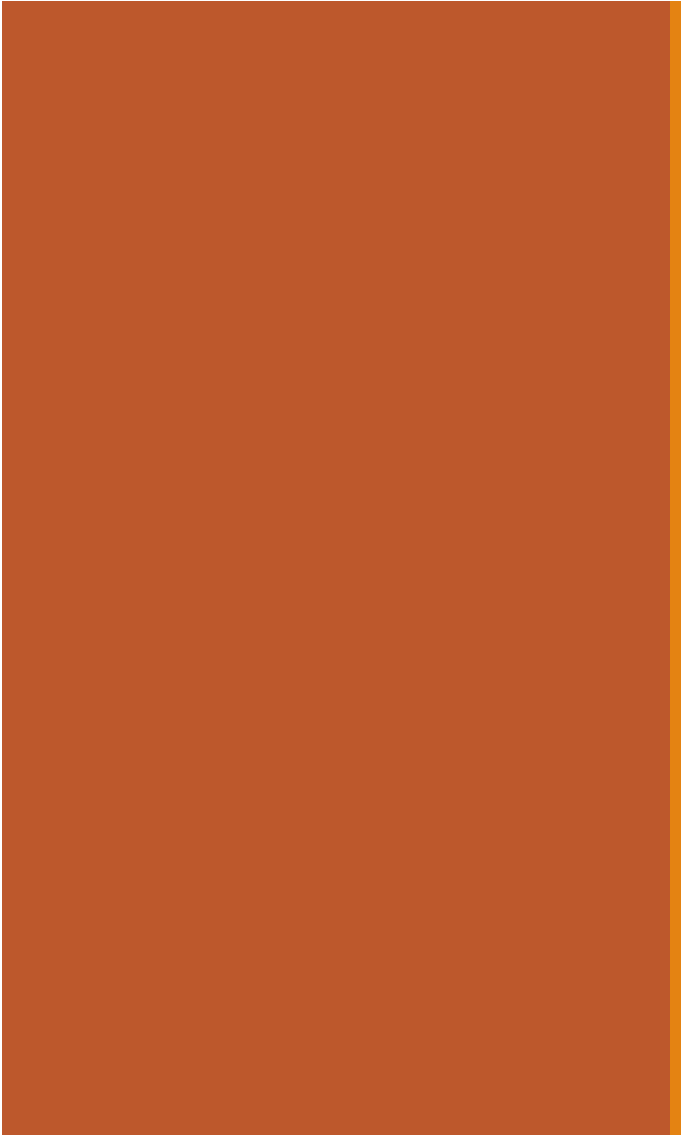
...

1.08



$$\bar{X}_2 = 2.4$$

Redux: The happiness of Nepal and Bhutan have a 0.7 difference of means, and this is statistically significant ($p < 0.05$).



Parameter Estimation

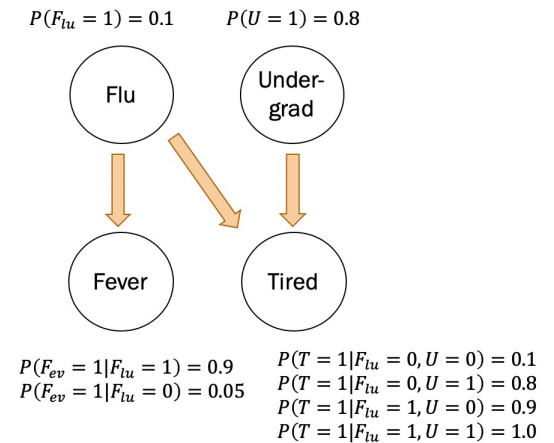
Story so far

At this point:

When provided with a **model** and **all the necessary parameters**, we can make predictions!

$$Y \sim \text{Exp}\left(\frac{1}{5}\right)$$

$$X_1, \dots, X_n \text{ iid}$$
$$X_i \sim \text{Ber}(0.2),$$
$$X = \sum_{i=1}^n X_i$$



Can we **infer** parameters for a given model when provided with some data?

~~What if you want to learn the **structure** of the model, too?~~

Glimpse: Week 10

Machine Learning

Some estimators

$X_1, X_2, \dots, X_n$ are n iid random variables,

where X_i drawn from distribution F with $E[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$.

in general, we never truly know these values, so we need to make educated guesses backed up by theory.

Sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

unbiased **estimate** of μ

these are from last time

Sample variance:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

unbiased **estimate** of σ^2

potentially useful estimates if trying to infer parameters of a Gaussian

What are parameters?

def Most random variables we've seen thus far are **parametric models**:

Distribution = model + parameter θ

ex The distribution $\text{Ber}(0.2) \rightarrow$ model is Bernoulli, parameter is $\theta = 0.2$.

For each of the distributions below, what is the parameter θ ?

1. $\text{Ber}(p)$ $\theta = p$
2. $\text{Poi}(\lambda)$
3. $\text{Uni}(\alpha, \beta)$
4. $\mathcal{N}(\mu, \sigma^2)$
5. $Y = mX + b$



What are parameters?

def Most random variables we've seen thus far are **parametric models**:

Distribution = model + parameter θ

ex The distribution $\text{Ber}(0.2) \rightarrow$ model is Bernoulli, parameter is $\theta = 0.2$.

For each of the distributions below, what is the parameter θ ?

1. $\text{Ber}(p)$ $\theta = p$
2. $\text{Poi}(\lambda)$ $\theta = \lambda$
3. $\text{Uni}(\alpha, \beta)$ $\theta = (\alpha, \beta)$
4. $\mathcal{N}(\mu, \sigma^2)$ $\theta = (\mu, \sigma^2)$
5. $Y = mX + b$ $\theta = (m, b)$
this will come up in machine learning in a few weeks

θ is the parameter of a distribution.
 θ can be a vector.

Why do we care?

In the real world, we don't know the true parameters.

- But we **observe data**: # times coin comes up heads, # requests for RydeShare per minute, # visitors to website per day, offer amount for that used bike you can't sell

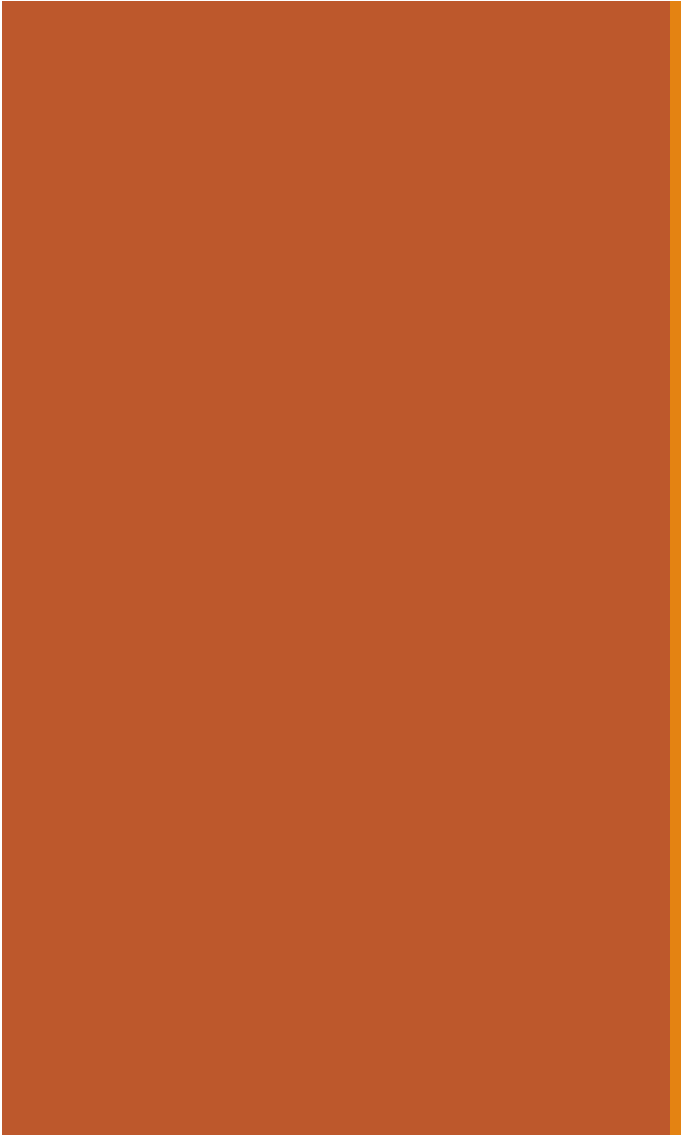
def **estimator** $\hat{\theta}$: a **random variable** estimating the true parameter θ .

whenever you see a carat over a parameter, it generally means it's an estimate

In parameter estimation,

We'll initially and often rely on **point estimates**—i.e., the best single value

- Provides an understanding of why data looks the way it does
- Can make future **predictions** using that model
- Can run simulations to generate more data



Maximum Likelihood Estimator

Defining the likelihood of data: Bernoulli

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

- X_i was drawn from distribution $F \sim \text{Ber}(\theta)$ with unknown parameter θ .
- Observed sample:

[0, 0, 1, 1, 1, 1, 1, 1, 1, 1]

($n = 10$)

*intuition tells us to
assume $\hat{p} = 0.8$, but
is that intuition correct?*

How likely is this sample if, say, $\theta = 0.4$?

*conditioned on the belief
that $\theta = p = 0.4$*

$$P(\text{sample} | \theta = 0.4) = (0.4)^8 (0.6)^2 = 0.000236$$

Likelihood of data
given parameter $\theta = 0.4$

Is there a better choice for θ ?

```
>>> for step in range(11):
...     theta = 0.1 * step
...     print("P(sample | theta = {:.1f}) = {:.1f}^8 * {:.1f}^2 = {:.6f}".format(theta, theta, 1 - theta, (theta ** 8) * ((1 - theta) ** 2)))
...
P(sample | theta = 0.0) = (0.0)^8 * (1.0)^2 = 0.000000
P(sample | theta = 0.1) = (0.1)^8 * (0.9)^2 = 0.000000
P(sample | theta = 0.2) = (0.2)^8 * (0.8)^2 = 0.000002
P(sample | theta = 0.3) = (0.3)^8 * (0.7)^2 = 0.000032
P(sample | theta = 0.4) = (0.4)^8 * (0.6)^2 = 0.000236
P(sample | theta = 0.5) = (0.5)^8 * (0.5)^2 = 0.000977
P(sample | theta = 0.6) = (0.6)^8 * (0.4)^2 = 0.002687
P(sample | theta = 0.7) = (0.7)^8 * (0.3)^2 = 0.005188
P(sample | theta = 0.8) = (0.8)^8 * (0.2)^2 = 0.006711
P(sample | theta = 0.9) = (0.9)^8 * (0.1)^2 = 0.004305
P(sample | theta = 1.0) = (1.0)^8 * (0.0)^2 = 0.000000
>>>
```

Defining the likelihood of data

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

- X_i was drawn from a distribution with density function $f(X_i|\theta)$.
- Sample: $(X_1, X_2, \dots, X_n)$

Likelihood question:

How likely is the sample $(X_1, X_2, \dots, X_n)$ given the parameter θ ?

Likelihood function, $L(\theta)$:

$$L(\theta) = f(X_1, X_2, \dots, X_n | \theta) = \prod_{i=1}^n f(X_i | \theta)$$

Handwritten notes:

- this is just a general definition of likelihood* (bracketed over the entire equation)
- X_i are iid, so by definition they are independent* (bracketed over the product)
- we even use f when the X_i are all discrete RVs.* (pointing to the f in the product)

This is just a product, since the X_i are iid.

Maximum Likelihood Estimator

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$, drawn from a distribution $f(X_i|\theta)$.

def The **Maximum Likelihood Estimator (MLE)** of θ is the value of θ that maximizes $L(\theta)$. *restated: θ_{MLE} is the value that maximizes the probability of the data you're observing.*

$$\theta_{MLE} = \arg \max_{\theta} L(\theta)$$

Maximum Likelihood Estimator

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$, drawn from a distribution $f(X_i|\theta)$.

def The **Maximum Likelihood Estimator (MLE)** of θ is the value of θ that maximizes $L(\theta)$.

$$\theta_{MLE} = \arg \max_{\theta} L(\theta)$$

Likelihood of your sample

$$L(\theta) = \prod_{i=1}^n f(X_i|\theta)$$

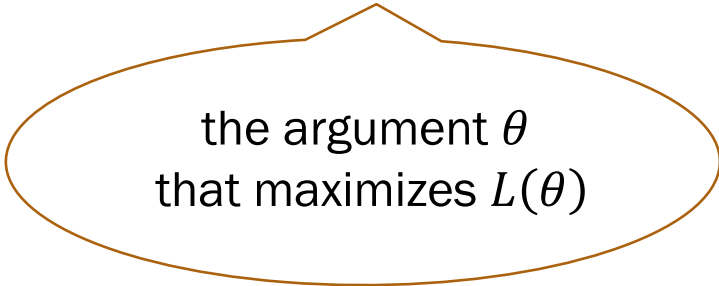
For continuous X_i , $f(X_i|\theta)$ is PDF, and for discrete X_i , $f(X_i|\theta)$ is PMF

Maximum Likelihood Estimator

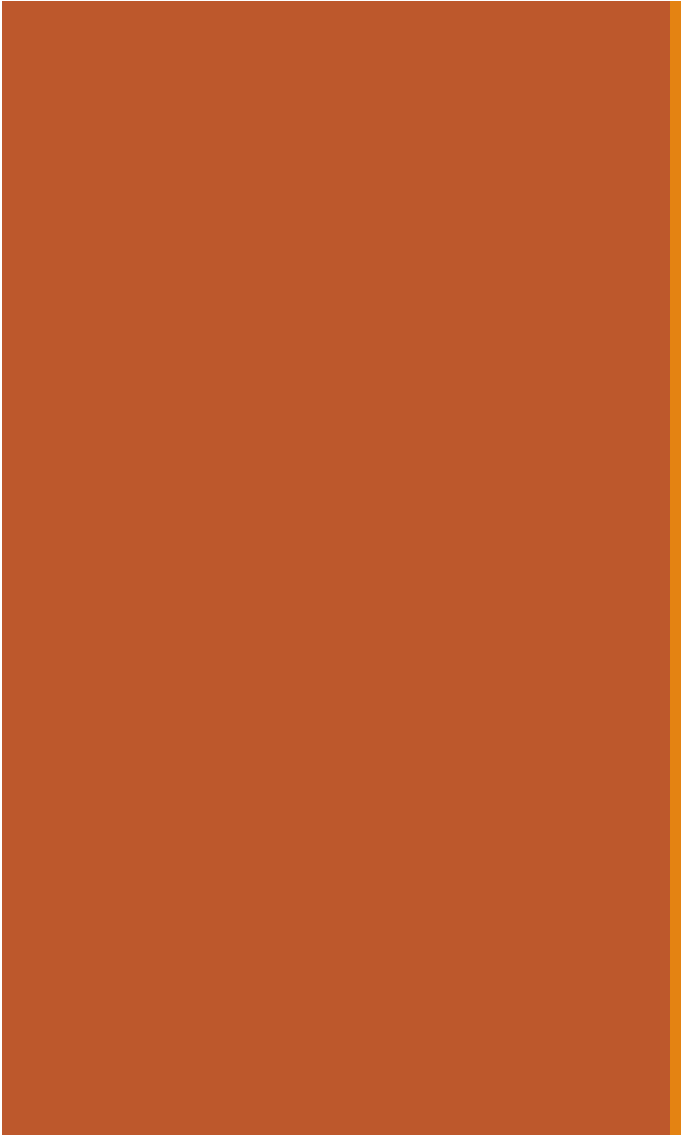
Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$, drawn from a distribution $f(X_i|\theta)$.

def The **Maximum Likelihood Estimator (MLE)** of θ is the value of θ that maximizes $L(\theta)$.

$$\theta_{MLE} = \arg \max_{\theta} L(\theta)$$



the argument θ
that maximizes $L(\theta)$



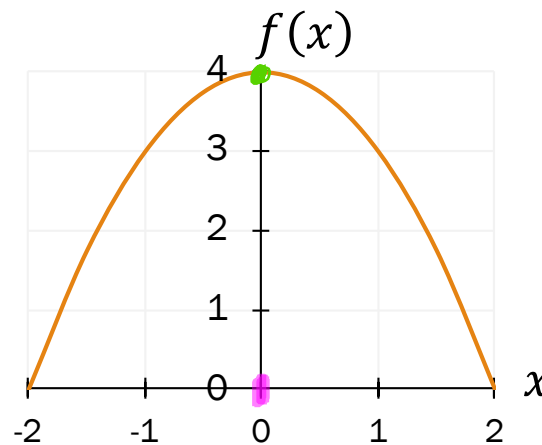
argmax and log
likelihood

New function: arg max

$$\arg \max_x f(x)$$

The argument x that maximizes the function $f(x)$.

Let $f(x) = -x^2 + 4$,
where $-2 < x < 2$.



1. $\max_x f(x) ?$

$= 4$

2. $\arg \max_x f(x) ?$

$= 0$

Argmax properties

$$\arg \max_x f(x)$$

The argument x that maximizes the function $f(x)$.

$$= \arg \max_x \log f(x)$$

(log is an increasing function:
 $x < y \Leftrightarrow \log x < \log y$)

$$= \arg \max_x (c \log f(x))$$

($x < y \Leftrightarrow c \log x < c \log y$)

for any positive constant c

Finding the argmax with calculus

$$\hat{x} = \arg \max_x f(x)$$

Let $f(x) = -x^2 + 4$,
where $-2 < x < 2$.

Differentiate w.r.t.
argmax's argument

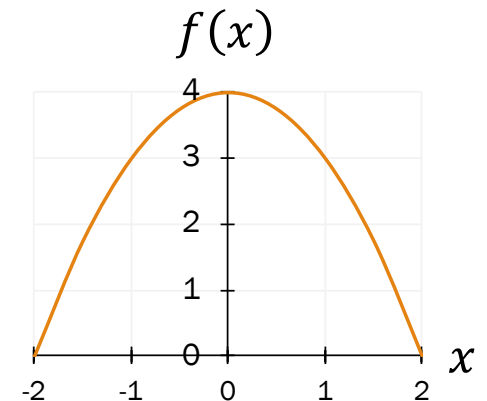
$$\frac{d}{dx} f(x) = \frac{d}{dx} (x^2 + 4) = 2x$$

Set to 0 and solve

$$2x = 0 \quad \Rightarrow \quad \hat{x} = 0$$

Make sure $\hat{x}$
is a maximum

- Check $f(\hat{x} \pm \epsilon) < f(\hat{x})$
- Often ignored in expository derivations
- We'll ignore it as well (and we won't require it in class or on problem sets and exams)



Maximum Likelihood Estimator

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$, drawn from a distribution $f(X_i|\theta)$.

$$L(\theta) = \prod_{i=1}^n f(X_i|\theta)$$

θ_{MLE} maximizes the likelihood of our sample, $L(\theta)$:

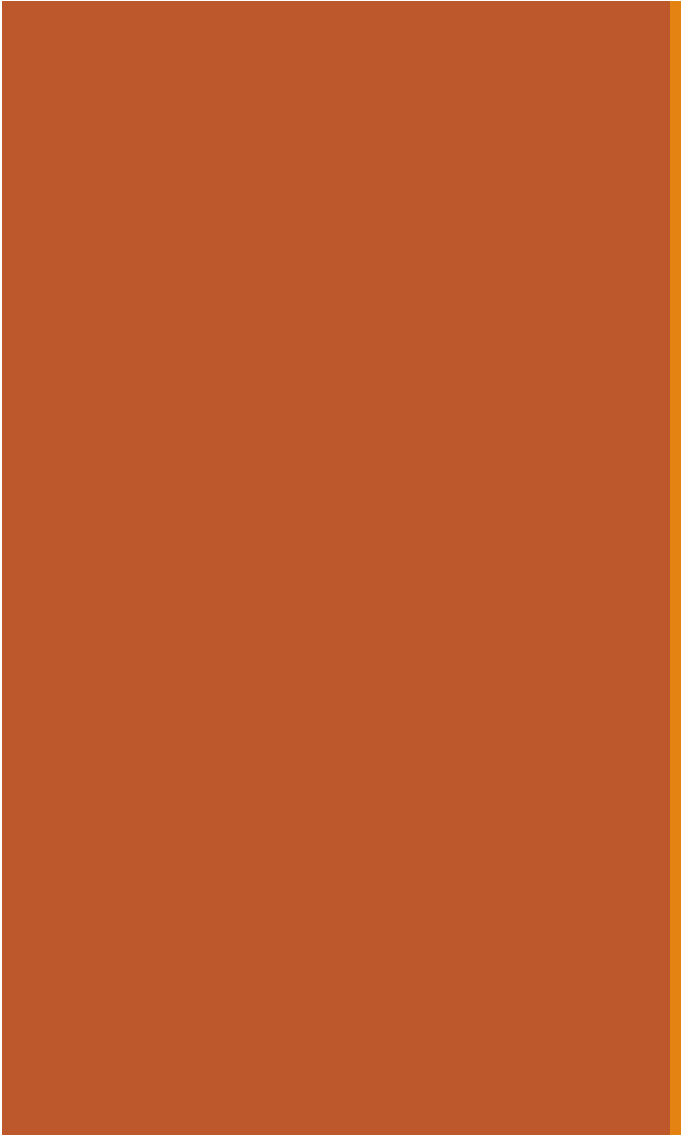
$$\theta_{MLE} = \arg \max_{\theta} L(\theta)$$

θ_{MLE} also maximizes the **log-likelihood function**, $LL(\theta)$:

$$\theta_{MLE} = \arg \max_{\theta} LL(\theta)$$

$$LL(\theta) = \log L(\theta) = \log \left(\prod_{i=1}^n f(X_i|\theta) \right) = \sum_{i=1}^n \log f(X_i|\theta)$$

$LL(\theta)$ is often easier to differentiate than $L(\theta)$.



MLE: Bernoulli

Computing the MLE

$$\theta_{MLE} = \arg \max_{\theta} LL(\theta)$$

General approach for finding θ_{MLE} , the MLE of θ :

1. Determine formula for $LL(\theta)$

$$LL(\theta) = \sum_{i=1}^n \log f(X_i|\theta)$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$\frac{\partial LL(\theta)}{\partial \theta}$$

3. Solve resulting equations

$$\text{To maximize: } \frac{\partial LL(\theta)}{\partial \theta} = 0$$

algebra or computer

$LL(\theta)$ is often easier to differentiate than $L(\theta)$.

Maximum Likelihood with Bernoulli

Consider a sample of n iid RVs $X_1, X_2, \dots, X_n$.

What is $\theta_{MLE} = p_{MLE}$?

- Let $X_i \sim \text{Ber}(p)$.

- Determine formula for $LL(\theta)$

$$LL(\theta) = \sum_{i=1}^n \log f(X_i|p)$$

$$f(X_i|p) = \begin{cases} p & \text{if } X_i = 1 \\ 1 - p & \text{if } X_i = 0 \end{cases}$$

- Differentiate $LL(\theta)$ wrt (each) θ , set to 0

- Solve resulting equations

$f(x_i|p)$, when expressed in this form, is not differentiable, so $\log f(x_i|p)$ isn't either



Maximum Likelihood with Bernoulli

Consider a sample of n iid RVs $X_1, X_2, \dots, X_n$.

What is $\theta_{MLE} = p_{MLE}$?

- Let $X_i \sim \text{Ber}(p)$.
- $f(X_i|p) = p^{X_i}(1-p)^{1-X_i}$

1. Determine formula for $LL(\theta)$

$$LL(\theta) = \sum_{i=1}^n \log f(X_i|p)$$

$$f(X_i|p) = \begin{cases} p & \text{if } X_i = 1 \\ 1-p & \text{if } X_i = 0 \end{cases}$$

2. Differentiate $LL(\theta)$ wrt (each) θ , set to 0

3. Solve resulting equations

this new, differentiable form is equivalent to the original for the two values of X_i we care about.

this is differentiable, but is it equivalent?

when $X_i = 1 \Rightarrow f(X_i=1|p) = p^1(1-p)^0 = p$ ✓

when $X_i = 0 \Rightarrow f(X_i=0|p) = p^0(1-p)^1 = 1-p$ ✓

✓

- differentiable with respect to p
- valid PMF over discrete domain

Maximum Likelihood with Bernoulli

Consider a sample of n iid RVs $X_1, X_2, \dots, X_n$.

What is $\theta_{MLE} = p_{MLE}$?

- Let $X_i \sim \text{Ber}(p)$.
- $f(X_i|p) = p^{X_i}(1-p)^{1-X_i}$

1. Determine formula for $LL(\theta)$

$$LL(\theta) = \sum_{i=1}^n \log f(X_i|p) = \sum_{i=1}^n \log p^{X_i}(1-p)^{1-X_i}$$
$$= \sum_{i=1}^n \left[\log p^{X_i} + \log (1-p)^{1-X_i} \right]$$

2. Differentiate $LL(\theta)$ wrt (each) θ , set to 0

$$= \sum_{i=1}^n [X_i \log p + (1 - X_i) \log(1 - p)]$$
$$= \log p \sum_{i=1}^n X_i + \log(1-p) \sum_{i=1}^n 1 - \log(1-p) \sum_{i=1}^n X_i$$

3. Solve resulting equations

$$= Y(\log p) + (n - Y) \log(1 - p), \text{ where } Y = \sum_{i=1}^n X_i$$

Maximum Likelihood with Bernoulli

Consider a sample of n iid RVs $X_1, X_2, \dots, X_n$.

What is $\theta_{MLE} = p_{MLE}$?

- Let $X_i \sim \text{Ber}(p)$.
- $f(X_i|p) = p^{X_i}(1-p)^{1-X_i}$

1. Determine formula for $LL(\theta)$

$$\begin{aligned} LL(\theta) &= \sum_{i=1}^n [X_i \log p + (1 - X_i) \log(1 - p)] \\ &= Y(\log p) + (n - Y) \log(1 - p), \text{ where } Y = \sum_{i=1}^n X_i \end{aligned}$$

2. Differentiate $LL(\theta)$ wrt (each) θ , set to 0

$$\frac{\partial LL(\theta)}{\partial p} = Y \frac{1}{p} + (n - Y) \frac{-1}{1 - p} = 0$$

3. Solve resulting equations

Maximum Likelihood with Bernoulli

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2. Differentiate $LL(\theta)$ wrt (each) θ , set to 0

$$\frac{\partial LL(\theta)}{\partial p} = Y \frac{1}{p} + (n - Y) \frac{-1}{1 - p} = 0 \Rightarrow \frac{Y}{p} = \frac{n - Y}{1 - p}$$

3. Solve resulting equations

$$Y - Yp = np - Yp$$
$$p = \frac{Y}{n} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

Maximum Likelihood with Bernoulli

Consider a sample of n iid RVs $X_1, X_2, \dots, X_n$.

What is $\theta_{MLE} = p_{MLE}$?

- Let $X_i \sim \text{Ber}(p)$.
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2. Differentiate $LL(\theta)$ wrt (each) θ , set to 0

$$\frac{\partial LL(\theta)}{\partial p} = Y \frac{1}{p} + (n - Y) \frac{-1}{1 - p} = 0$$

3. Solve resulting equations

$$p_{MLE} = \frac{1}{n} Y = \frac{1}{n} \sum_{i=1}^n X_i$$

MLE of the Bernoulli parameter, p_{MLE} , is the sample mean, $\bar{X}$, which is an unbiased estimator of the true mean.

Quick check

- You draw n iid random variables $X_1, X_2, \dots, X_n$ from the distribution F , yielding the following sample:

$[0, 0, 1, 1, 1, 1, 1, 1, 1, 1]$ $(n = 10)$

- Suppose distribution $F = \text{Ber}(p)$ with unknown parameter p .

1. What is p_{MLE} , the MLE of the parameter p ?

- A. 1.0
- B. 0.5
- ☒ C. 0.8
- D. 0.2
- E. None/other

$$p_{MLE} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$



Quick check

- You draw n iid random variables $X_1, X_2, \dots, X_n$ from the distribution F , yielding the following sample:

$$[0, 0, 1, 1, 1, 1, 1, 1, 1, 1] \quad (n = 10)$$

- Suppose distribution $F = \text{Ber}(p)$ with unknown parameter p .

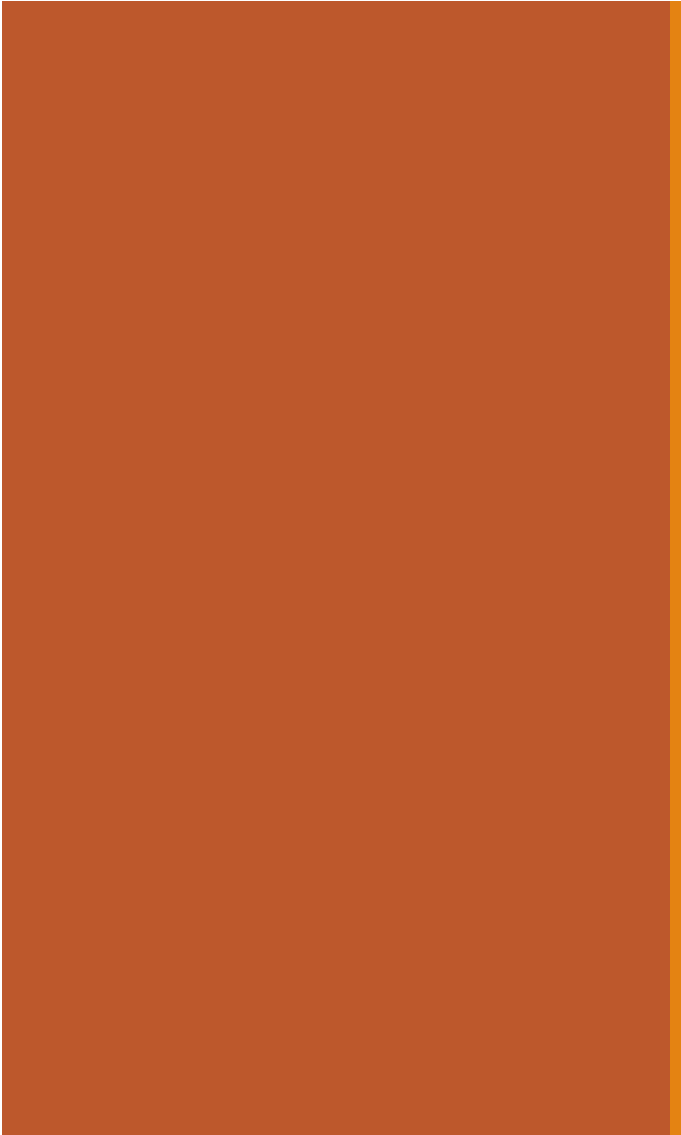
- What is p_{MLE} , the MLE of the parameter p ? C. 0.8
- What is the likelihood $L(\theta)$ of this specific sample?

$$f(X_i|p) = p^{X_i}(1-p)^{1-X_i} \text{ where } X_i \in \{0,1\}$$

$$L(\theta) = \prod_{i=1}^n f(X_i|p) \quad \text{where } \theta = p$$

$$= p^8(1-p)^2 = 0.8^8 0.2^2 = 0.0067$$

yes, it's small
but any specific
sequence of 10
Bernoulli trials
is going to be
small.



MLE: Poisson and Uniform

Maximum Likelihood with Poisson

Consider a sample of n iid RVs $X_1, X_2, \dots, X_n$.

What is $\theta_{MLE} = \lambda_{MLE}$? *reminder: $\log ab = \log a + \log b$
 $\log \frac{a}{b} = \log a - \log b$*

- Let $X_i \sim \text{Poi}(\lambda)$.
- PMF: $f(X_i|\lambda) = \frac{e^{-\lambda} \lambda^{X_i}}{X_i!}$

1. Determine
formula for $LL(\theta)$

$$\begin{aligned} LL(\theta) &= \sum_{i=1}^n \log \left(\frac{e^{-\lambda} \lambda^{X_i}}{X_i!} \right) = \sum_{i=1}^n (-\lambda \log e + X_i \log \lambda - \log X_i!) \\ &= -n\lambda + \log(\lambda) \sum_{i=1}^n X_i - \sum_{i=1}^n \log(X_i!) \end{aligned}$$

using natural log,
i.e., $\ln e = 1$

Maximum Likelihood with Poisson

Consider a sample of n iid RVs $X_1, X_2, \dots, X_n$.

What is $\theta_{MLE} = \lambda_{MLE}$?

- Let $X_i \sim \text{Poi}(\lambda)$.
- PMF: $f(X_i|\lambda) = \frac{e^{-\lambda} \lambda^{X_i}}{X_i!}$

1. Determine formula for $LL(\theta)$

$$LL(\theta) = \sum_{i=1}^n \log \left(\frac{e^{-\lambda} \lambda^{X_i}}{X_i!} \right) = \sum_{i=1}^n (-\lambda \log e + X_i \log \lambda - \log X_i!)$$

$$= -n\lambda + \log(\lambda) \sum_{i=1}^n X_i - \sum_{i=1}^n \log(X_i!)$$

using natural log,
i.e., $\ln e = 1$ zero

2. Differentiate $LL(\theta)$
w.r.t. (each) θ , set to 0

$$\frac{\partial LL(\theta)}{\partial \lambda} = ?$$

$$\frac{d}{d\lambda} (-n\lambda) + \frac{d}{d\lambda} \log \lambda \sum_{i=1}^n X_i - \frac{d}{d\lambda} \sum_{i=1}^n \log(X_i!)$$

$-n + \frac{1}{\lambda} \sum_{i=1}^n X_i$

A.
$$-n + \frac{1}{\lambda} \sum_{i=1}^n X_i + n \log \lambda - \sum_{i=1}^n \frac{1}{X_i!} \cdot \frac{\partial X_i!}{\partial X_i}$$

B.
$$-n + \frac{1}{\lambda} \sum_{i=1}^n X_i$$

C. Stop trying



Maximum Likelihood with Poisson

Consider a sample of n iid RVs $X_1, X_2, \dots, X_n$.

What is $\theta_{MLE} = \lambda_{MLE}$?

- Let $X_i \sim \text{Poi}(\lambda)$.
- PMF: $f(X_i|\lambda) = \frac{e^{-\lambda} \lambda^{X_i}}{X_i!}$

1. Determine formula for $LL(\theta)$

$$\begin{aligned} LL(\theta) &= \sum_{i=1}^n \log \left(\frac{e^{-\lambda} \lambda^{X_i}}{X_i!} \right) = \sum_{i=1}^n (-\lambda \log e + X_i \log \lambda - \log X_i!) \\ &= -n\lambda + \log(\lambda) \sum_{i=1}^n X_i - \sum_{i=1}^n \log(X_i!) \end{aligned}$$

using natural log,
i.e., $\ln e = 1$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ , set to 0

$$\frac{\partial LL(\theta)}{\partial \lambda} = -n + \frac{1}{\lambda} \sum_{i=1}^n X_i = 0 \Rightarrow \frac{1}{\lambda} \sum_{i=1}^n X_i = n$$

3. Solve resulting equations

$$\lambda_{MLE} = \frac{1}{n} \sum_{i=1}^n X_i$$

MLE of the Poisson parameter, λ_{MLE} , is the sample mean, $\bar{X}$, which is an unbiased estimator of the true mean.

Quick Review

1. A particular experiment can be modeled as a Poisson RV with parameter λ , in terms of events/minute. sample is $(X_1 = x_1, X_2 = x_2, \dots, X_{10} = x_{10})$
Collect data: observe 53 events over the next 10 minutes. What is λ_{MLE} ?

$$\lambda_{MLE} = \frac{1}{n} \sum_{i=1}^n X_i$$
$$\sum_{i=1}^{10} X_i = 53 \Rightarrow \lambda_{MLE} = \frac{1}{10} 53 = 5.3$$

2. Is the Bernoulli MLE an unbiased estimator of the Bernoulli parameter p ? ☒

$$X \sim \text{Ber}(p) \quad E[p_{MLE}] = p?$$
$$E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = E[\bar{X}] = p$$

3. Is the Poisson MLE an unbiased estimator of the Poisson variance? ☒

$$X \sim \text{Poi}(\lambda) \quad E[\lambda_{MLE}] = E[\bar{X}] = \lambda = \sigma^2$$

4. What does unbiased mean?

$$E[\text{estimator}] = \text{the truth}$$

Unbiased: If you repeat your experiment multiple times, on average, you'll get what you are looking for.



Maximum Likelihood with Uniform

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

Let $X_i \sim \text{Uni}(\alpha, \beta)$.

$$f(X_i|\alpha, \beta) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } \alpha \leq x_i \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

1. Determine formula for $L(\theta)$

$$L(\theta) = \begin{cases} \left(\frac{1}{\beta - \alpha}\right)^n & \text{if } \alpha \leq x_1, x_2, \dots, x_n \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

$LL(\theta) = n \log \frac{1}{\beta - \alpha}$ for $\alpha \leq x_1, x_2, \dots, x_n \leq \beta$

2. Differentiate $L(\theta)$ wrt each θ , set to 0

- A. Great, let's do it
- B. Use $LL(\theta)$ instead
- C.** Constraint $\alpha \leq x_1, x_2, \dots, x_n \leq \beta$ makes differentiation hard



Maximum Likelihood with Uniform: Sample

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

Let $X_i \sim \text{Uni}(\alpha, \beta)$.

$$L(\theta) = \begin{cases} \left(\frac{1}{\beta - \alpha}\right)^n & \text{if } \alpha \leq x_1, x_2, \dots, x_n \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

Underlying $X_i \sim \text{Uni}(0,1)$ $[0.15, 0.20, 0.30, 0.40, 0.65, 0.70, 0.75]$

You observe data:

Which parameters
maximize $L(\theta)$?

- A. $\text{Uni}(\alpha = 0.00, \beta = 1.00)$
- B. $\text{Uni}(\alpha = 0.15, \beta = 0.75)$
- C. $\text{Uni}(\alpha = 0.15, \beta = 0.70)$



Maximum Likelihood with Uniform: Sample

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

Let $X_i \sim \text{Uni}(\alpha, \beta)$.

$$L(\theta) = \begin{cases} \left(\frac{1}{\beta - \alpha}\right)^n & \text{if } \alpha \leq x_1, x_2, \dots, x_n \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

Underlying $X_i \sim \text{Uni}(0,1)$ $[0.15, 0.20, 0.30, 0.40, 0.65, 0.70, 0.75]$

You observe data:

Which parameters maximize $L(\theta)$?

A. $\text{Uni}(\alpha = 0.00, \beta = 1.00)$

$$(1)^7 = 1$$

this is the likelihood using "unknown" underlying parameters

B. $\text{Uni}(\alpha = 0.15, \beta = 0.75)$

$$\left(\frac{1}{0.6}\right)^7 = 59.5$$

C. $\text{Uni}(\alpha = 0.15, \beta = 0.70)$

$$\left(\frac{1}{0.55}\right)^6 \cdot 0 = 0$$

included on behalf of the 0.75



Original parameters may not yield maximum likelihood.

Maximum Likelihood with Uniform

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

Let $X_i \sim \text{Uni}(\alpha, \beta)$.

$$L(\theta) = \begin{cases} \left(\frac{1}{\beta - \alpha}\right)^n & \text{if } \alpha \leq x_1, x_2, \dots, x_n \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

$$\theta_{MLE}: \alpha_{MLE} = \min(x_1, x_2, \dots, x_n) \quad \beta_{MLE} = \max(x_1, x_2, \dots, x_n)$$

Intuition:

- Want interval size $\beta - \alpha$ to be as narrow as possible to maximize likelihood function.
- Need to ensure all datapoints are included in interval. Otherwise, $L(\theta) = 0$.

([demo](#))

Small samples = problems with MLE

Maximum Likelihood Estimator θ_{MLE} :

$$\theta_{MLE} = \arg \max_{\theta} L(\theta)$$

- Best explains the data we've seen
- Does not attempt to generalize to data not yet observed.



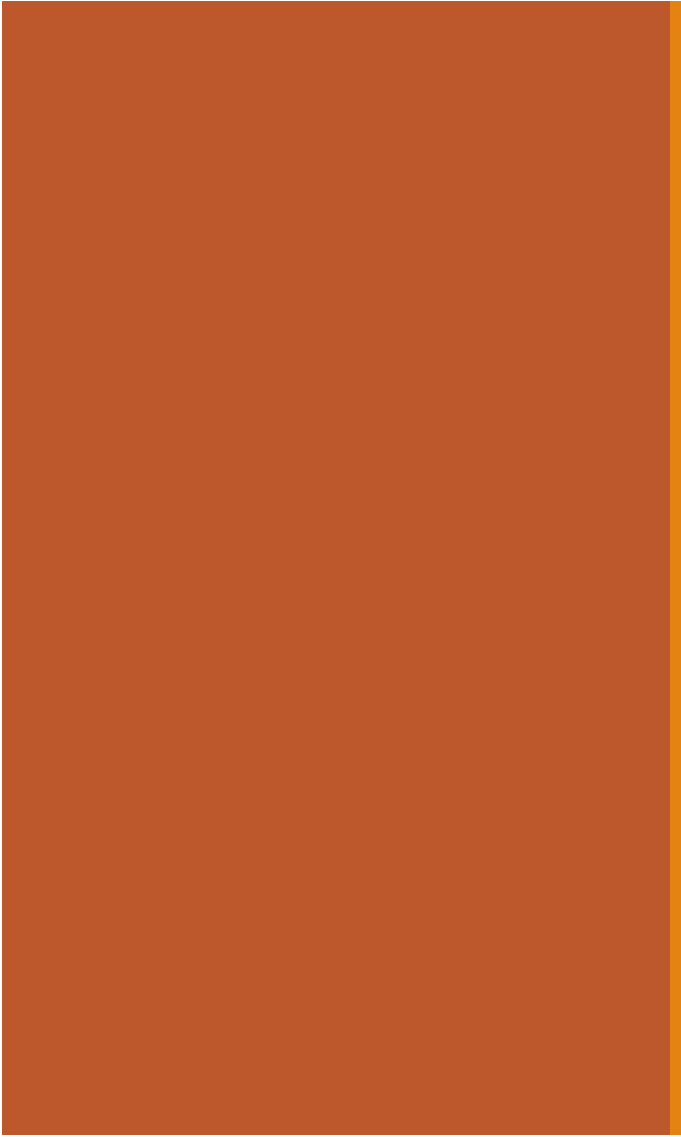
In many cases, $\mu_{MLE} = \frac{1}{n} \sum_{i=1}^n X_i$ Sample mean (MLE for Bernoulli p , Poisson λ , Normal μ)

- Unbiased ($E[\mu_{MLE}] = \mu$, regardless of sample size)



For some cases, like Uniform: $\alpha_{MLE} \geq \alpha$, $\beta_{MLE} \leq \beta$

- Ad hoc, biased, and problematic for small sample sizes
- Example: If $n = 1$, then $\alpha = \beta$, and our estimates yield an invalid distribution



MLE: Gaussian

Maximum Likelihood with Normal

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

- Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$.

$$f(X_i | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(X_i - \mu)^2 / (2\sigma^2)}$$

What is $\theta_{MLE} = (\mu_{MLE}, \sigma_{MLE}^2)$?

*ack!
two parameters*

1. Determine
formula for $LL(\theta)$

2. Differentiate $LL(\theta)$
wrt (each) θ , set to 0

3. Solve resulting
equations

$$\begin{aligned} LL(\theta) &= \sum_{i=1}^n \log \left(\frac{1}{\sqrt{2\pi}\sigma} e^{-(X_i - \mu)^2 / (2\sigma^2)} \right) = \sum_{i=1}^n [-\log(\sqrt{2\pi}\sigma) - (X_i - \mu)^2 / (2\sigma^2)] \\ &= -\sum_{i=1}^n \log(\sqrt{2\pi}\sigma) - \sum_{i=1}^n [(X_i - \mu)^2 / (2\sigma^2)] \end{aligned}$$

(using natural log)

Maximum Likelihood with Normal

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

- Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$.

$$f(X_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(X_i - \mu)^2 / (2\sigma^2)}$$

What is $\theta_{MLE} = (\mu_{MLE}, \sigma_{MLE}^2)$?

1. Determine formula for $LL(\theta)$
2. Differentiate $LL(\theta)$ wrt (each) θ , set to 0
3. Solve resulting equations

with respect to μ

$$LL(\theta) = - \sum_{i=1}^n \log(\sqrt{2\pi}\sigma) - \sum_{i=1}^n [(X_i - \mu)^2 / (2\sigma^2)]$$

$$\frac{\partial LL(\theta)}{\partial \mu} = \sum_{i=1}^n [2(X_i - \mu) / (2\sigma^2)]$$

$$= \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu) = 0$$

Maximum Likelihood with Normal

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

- Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$.

$$f(X_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(X_i - \mu)^2 / (2\sigma^2)}$$

What is $\theta_{MLE} = (\mu_{MLE}, \sigma_{MLE}^2)$?

1. Determine
formula for $LL(\theta)$

2. Differentiate $LL(\theta)$
wrt (each) θ , set to 0

3. Solve resulting
equations

with respect to μ $LL(\theta) = -\sum_{i=1}^n \log(\sqrt{2\pi}\sigma) - \sum_{i=1}^n [(X_i - \mu)^2 / (2\sigma^2)]$ with respect to σ

$$\frac{\partial LL(\theta)}{\partial \mu} = \sum_{i=1}^n [2(X_i - \mu) / (2\sigma^2)]$$

$$= \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu) = 0$$

$$\frac{\partial LL(\theta)}{\partial \sigma} = -\sum_{i=1}^n \frac{1}{\sigma} + \sum_{i=1}^n 2(X_i - \mu)^2 / (2\sigma^3)$$

$$= -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (X_i - \mu)^2 = 0$$

Maximum Likelihood with Normal

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

- Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$.

$$f(X_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(X_i - \mu)^2 / (2\sigma^2)}$$

What is $\theta_{MLE} = (\mu_{MLE}, \sigma_{MLE}^2)$?

3. Solve resulting equations

Two equations, two unknowns:

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu) = 0$$

$$-\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (X_i - \mu)^2 = 0$$

First, solve for μ_{MLE} :

$$\frac{1}{\sigma^2} \sum_{i=1}^n X_i - \frac{1}{\sigma^2} \sum_{i=1}^n \mu = 0$$

$$\Rightarrow \sum_{i=1}^n X_i = n\mu$$

$$\Rightarrow \mu_{MLE} = \frac{1}{n} \sum_{i=1}^n X_i$$

unbiased

another sample mean

Maximum Likelihood with Normal

Consider a sample of n iid random variables $X_1, X_2, \dots, X_n$.

- Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$.

$$f(X_i|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(X_i - \mu)^2 / (2\sigma^2)}$$

What is $\theta_{MLE} = (\mu_{MLE}, \sigma_{MLE}^2)$?

3. Solve resulting equations

Two equations,
two unknowns:

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu) = 0$$

$$-\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (X_i - \mu)^2 = 0$$

First, solve
for μ_{MLE} :

$$\frac{1}{\sigma^2} \sum_{i=1}^n X_i - \frac{1}{\sigma^2} \sum_{i=1}^n \mu = 0 \Rightarrow \sum_{i=1}^n X_i = n\mu \Rightarrow \mu_{MLE} = \frac{1}{n} \sum_{i=1}^n X_i$$

unbiased

Next, solve
for σ_{MLE} :

$$\frac{1}{\sigma^3} \sum_{i=1}^n (X_i - \mu)^2 = \frac{n}{\sigma} \Rightarrow \sum_{i=1}^n (X_i - \mu)^2 = \sigma^2 n \Rightarrow \sigma_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu_{MLE})^2$$

biased !!